PROBABILITY PROBLEM SHEET

This sheet covers the lectures on probability, on interpreting experimental results, and on Bayesian inference. Question 1 practises the rules of probability, Question 2 error bars and significance tests, and Question 3 Bayesian updating. Each ends with a short Python part (`import numpy as np` is assumed). Question 4 is harder and optional: it does Bayesian updating for normal distributions and for linear regression.

1

(a) Events A and B have $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.5$ and $\mathbb{P}(A \cap B) = 0.1$. Find $\mathbb{P}(A \cup B)$, $\mathbb{P}(\neg A)$ and $\mathbb{P}(A \mid B)$. Are A and B independent?

Adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts $A \cap B$ twice, so $\mathbb{P}(A \cup B) = 0.3 + 0.5 - 0.1 = 0.7$. $\mathbb{P}(\neg A) = 1 - 0.3 = 0.7$. $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) = 0.1/0.5 = 0.2$. They are not independent: $\mathbb{P}(A)\mathbb{P}(B) = 0.15 \neq 0.1$ (equivalently $\mathbb{P}(A \mid B) = 0.2 \neq \mathbb{P}(A)$).

[4 marks]

(b) The joint distribution of the weather W and whether I carry an umbrella U is

	$U =$ yes	$U =$ no
$W =$ sun	0.05	0.55
$W =$ rain	0.25	0.15

Find $\mathbb{P}(U = \text{yes})$, $\mathbb{P}(W = \text{rain} \mid U = \text{yes})$ and $\mathbb{P}(U = \text{yes} \mid W = \text{rain})$. Comment on the last two.

Marginalising, $\mathbb{P}(U = \text{yes}) = 0.05 + 0.25 = 0.30$ and $\mathbb{P}(W = \text{rain}) = 0.25 + 0.15 = 0.40$. Then $\mathbb{P}(\text{rain} \mid \text{yes}) = 0.25/0.30 \approx 0.83$ and $\mathbb{P}(\text{yes} \mid \text{rain}) = 0.25/0.40 = 0.625$. The two conditional probabilities are different: conditioning is not symmetric, and the umbrella does not cause the rain.

[5 marks]

(c) X is the score on a fair dice. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ and $\mathrm{Var}(X)$. Use linearity of expectation to find $\mathbb{E}[2X + 3]$, and find $\mathrm{Var}(2X + 3)$.

$\mathbb{E}[X] = \frac{1}{6}(1 + \cdots + 6) = 3.5$ and $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + \cdots + 36) = \frac{91}{6}$, so $\mathrm{Var}(X) = \frac{91}{6} - 3.5^2 = \frac{35}{12}$. Then $\mathbb{E}[2X + 3] = 2 \times 3.5 + 3 = 10$, and since adding a constant does not change the spread, $\mathrm{Var}(2X + 3) = 4\mathrm{Var}(X) = \frac{35}{3}$.

[4 marks]

(d) X is uniform on $(0,1)$ and $Y = -\log(X)/\lambda$ with $\lambda > 0$. Use the change of variables formula $f_Y(y) = f_X(x)|\mathrm{d}x/\mathrm{d}y|$ to find the density of Y.

$f_X(x) = 1$ on $(0,1)$, and $x = \mathrm{e}^{-\lambda y}$ so $|\mathrm{d}x/\mathrm{d}y| = \lambda \mathrm{e}^{-\lambda y}$. Hence $f_Y(y) = \lambda \mathrm{e}^{-\lambda y}$ for $y > 0$: the exponential distribution. This is how computers turn uniform random numbers into exponential ones.

[4 marks]

(e) In Python, draw 100000 samples of Y from part (d) with $\lambda = 2$. Check that their mean is close to $1/\lambda$, and estimate $\mathbb{P}(Y \leq 0.5)$ as the average of the indicator $[[Y \leq 0.5]]$. Compare with the exact value $1 - \mathrm{e}^{-\lambda/2}$.

```
rng = np.random.default_rng(0)
lam = 2.0
y = -np.log(rng.uniform(size=100000)) / lam
print(y.mean(), 1/lam)
print(np.mean(y <= 0.5), 1 - np.exp(-lam*0.5))
```

The mean is 0.500 and the estimate 0.633, against 0.632: a probability is the expectation of an indicator.

[3 marks]

End of question 1

2

(a) A classifier gets 1700 of 2000 test examples right. Estimate its accuracy and standard error, and write the result down sensibly.

$\hat{p} = 0.85$ and $\Delta = \sqrt{0.85 \times 0.15/2000} \approx 0.008$, so the accuracy is 0.850 ± 0.008 (or $(85.0 \pm 0.8)\%$).

[3 marks]

(b) A second classifier gets 0.866 ± 0.008 on a *different* test set of 2000 examples. Is it significantly better at the 5% level?

The gap is 0.016. The two estimates are independent, so the standard error of the difference is $\sqrt{2} \times 0.008 \approx 0.0113$, not 0.008. Then $t \approx 1.42$ and, with so many examples, $p \approx 0.16$: not significant. The error bars are about two standard errors apart, which is not enough.

[4 marks]

(c) Now both classifiers are run on the *same* 2000 examples. The first is right and the second wrong on 20 examples; the second is right and the first wrong on 52; they agree on the rest. Let D_i be $+1$, -1 or 0 as the second classifier does better, worse or the same on example i. Use a paired t-test on the D_i. Why does the answer differ from part (b)?

$\bar{D} = (52 - 20)/2000 = 0.016$, the same gap as before. The sample variance of the D_i is about $72/2000 - 0.016^2 \approx 0.0357$, so the standard error is $\sqrt{0.0357/2000} \approx 0.0042$ and $t \approx 3.8$, giving $p \approx 0.0002$: highly significant. On the same examples most of the variation (easy and hard examples) is shared and cancels in the differences; only the 72 examples where they disagree carry information.

[5 marks]

(d) You try 20 hyperparameter settings, none of which is really any better than the baseline, and test each at the 5% level. What is the probability that at least one looks significant?

$1 - 0.95^{20} \approx 0.64$. Running many tests and reporting the best is cheating, even if it is done without meaning to.

[3 marks]

Answers

(e) Check part (b) in Python by simulating the null hypothesis: two classifiers that are both right with probability 0.85, each tested on its own 2000 examples. Repeat 100000 times and find the fraction of trials in which the accuracies differ by at least 0.016.

```
rng = np.random.default_rng(0)
a = rng.binomial(2000, 0.85, size=100000) / 2000
b = rng.binomial(2000, 0.85, size=100000) / 2000
print(np.mean(np.abs(a - b) >= 0.016))
```

This gives about 0.16, agreeing with part (b) (the small excess over 0.157 is because the accuracies can only take multiples of $1/2000$). Simulating the null hypothesis works even when no formula is available.

[5 marks]

End of question 2

3

(a) A disease affects 1% of people. A test detects it in 95% of people who have it, but is also positive for 5% of people who do not. Someone tests positive. What is the probability that they have the disease?

By Bayes' rule

$$\mathbb{P}(D \mid +) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} = \frac{0.0095}{0.0590} \approx 0.16.$$

Most positives are false positives, because the disease is rare: the prior matters.

[4 marks]

(b) A coin with unknown probability p of a head is tossed 10 times, giving 7 heads. Starting from the uniform prior $\mathrm{Beta}(p \mid 1,1)$, what is the posterior? Find its mean and its most probable value, and compare with the frequency $7/10$.

The posterior is $\mathrm{Beta}(p \mid 1+7, 1+3) = \mathrm{Beta}(p \mid 8,4)$. Its mean is $8/12 \approx 0.67$ and its mode $(8-1)/(8+4-2) = 0.7$, the frequency. The mean is pulled slightly towards $\frac{1}{2}$ by the prior.

[4 marks]

(c) Show that updating after each toss, using each posterior as the prior for the next toss, gives the same posterior as part (b), whatever the order of the heads and tails.

Each head adds 1 to the first parameter of the beta distribution and each tail adds 1 to the second. After all ten tosses we have added 7 and 3 in some order, giving $\mathrm{Beta}(p \mid 8,4)$ whatever the order.

[3 marks]

(d) The number of cars passing in an hour is Poisson with rate μ, and the prior is $\mathrm{Gamma}(\mu \mid 2,1)$. On three days we count 3, 5 and 4 cars. Using the update $a \to a+N$, $b \to b+1$ for each count, find the posterior and its mean and standard deviation (a/b and $\sqrt{a}/b$).

$a = 2+3+5+4 = 14$ and $b = 1+3 = 4$, so the posterior is $\mathrm{Gamma}(\mu \mid 14,4)$ with mean 3.5 and standard deviation $\sqrt{14}/4 \approx 0.94$.

[4 marks]

(e) In Python, simulate 100 tosses of a coin with $p = 0.7$ and update a $\mathrm{Beta}(1,1)$ prior after each toss. Print the posterior mean and standard deviation, $\sqrt{ab/((a+b)^2(a+b+1))}$, after 10, 30 and 100 tosses.

```
rng = np.random.default_rng(1)
a, b = 1, 1
for n, head in enumerate(rng.random(100) < 0.7, start=1):
    a += head; b += 1 - head
    if n in (10, 30, 100):
        print(n, a/(a+b), np.sqrt(a*b/((a+b)**2*(a+b+1))))
```

With this seed the means are about 0.67, 0.66 and 0.67 and the standard deviations 0.13, 0.08 and 0.05: the uncertainty shrinks roughly as $1/\sqrt{n}$.

[5 marks]

End of question 3

4 This question is harder and optional. When both the likelihood and the prior are normal, so is the posterior, and we only need to keep track of a mean and a variance (or a covariance matrix). A helpful way to read the prior is as *pseudo-data*: observations we pretend we have already seen.

(a) We measure an unknown quantity μ n times, with independent errors: $X_i \sim \mathcal{N}(\mu,\sigma^2)$ with σ known. Our prior is $\mu \sim \mathcal{N}(\mu_0,s_0^2)$. Write down the log of the posterior, up to a constant, and show that it is quadratic in μ. Hence show the posterior is $\mathcal{N}(\mu_n,s_n^2)$ with

$$\frac{1}{s_n^2} = \frac{1}{s_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = s_n^2\left(\frac{\mu_0}{s_0^2} + \frac{1}{\sigma^2}\sum_{i=1}^{n} X_i\right).$$

(Hint: the log of $\mathcal{N}(\mu \mid m,s^2)$ is $-\mu^2/(2s^2) + \mu m/s^2 +$ a constant; match the coefficients of μ^2 and μ.)

$$\begin{aligned}\log f(\mu \mid \mathcal{D}) &= -\frac{(\mu-\mu_0)^2}{2s_0^2} - \sum_{i=1}^{n}\frac{(X_i-\mu)^2}{2\sigma^2} + \text{const} \\ &= -\frac{\mu^2}{2}\left(\frac{1}{s_0^2}+\frac{n}{\sigma^2}\right) + \mu\left(\frac{\mu_0}{s_0^2} + \frac{\sum_i X_i}{\sigma^2}\right) + \text{const.}\end{aligned}$$

Matching the coefficient of μ^2 with $-1/(2s_n^2)$ and the coefficient of μ with μ_n/s_n^2 gives the result. Inverse variances (*precisions*) add.

[5 marks]

(b) Let $n_0 = \sigma^2/s_0^2$ and $\bar{X} = \frac{1}{n}\sum_i X_i$. Show that $\mu_n = (n_0\mu_0 + n\bar{X})/(n_0+n)$ and $s_n^2 = \sigma^2/(n_0+n)$. Interpret n_0. What happens when $s_0 \to \infty$, and when $n \to \infty$?

Multiplying top and bottom by σ^2, $1/s_n^2 = (n_0+n)/\sigma^2$ and $\mu_n = (n_0\mu_0 + n\bar{X})/(n_0+n)$. The prior acts like n_0 extra measurements, all equal to μ_0: a confident prior (small s_0) is worth many measurements. As $s_0 \to \infty$, $n_0 \to 0$, so $\mu_n = \bar{X}$ and $s_n = \sigma/\sqrt{n}$, the standard error from the lecture on experimental results. As $n \to \infty$ the data swamp the prior.

[4 marks]

(c) Take $\mu_0 = 0$, $s_0 = 2$, $\sigma = 1$ and measurements 1.2, 0.8, 1.5, 0.9. Find n_0, μ_n and s_n.

$n_0 = 1/4$, $n = 4$ and $\sum_i X_i = 4.4$, so $\mu_n = 4.4/4.25 \approx 1.04$ and $s_n = 1/\sqrt{4.25} \approx 0.49$. The prior pulls the mean $\bar{X} = 1.1$ slightly towards 0.

[3 marks]

(d) Now consider linear regression, $y_k = \boldsymbol{w}^\mathsf{T}\boldsymbol{\phi}_k + \epsilon_k$ with $\epsilon_k \sim \mathcal{N}(0,\sigma^2)$, where the feature vectors $\boldsymbol{\phi}_k$ are the rows of a matrix $\boldsymbol{\Phi}$. Take the prior $\boldsymbol{w} \sim \mathcal{N}(\mathbf{0},(\sigma^2/\eta)\mathbf{I})$: the weights are probably small. Using the multivariate version of the hint, $\mathrm{e}^{-\frac{1}{2}\boldsymbol{w}^\mathsf{T}\mathbf{A}\boldsymbol{w}+\boldsymbol{b}^\mathsf{T}\boldsymbol{w}} \propto \mathcal{N}(\boldsymbol{w} \mid \mathbf{A}^{-1}\boldsymbol{b},\mathbf{A}^{-1})$ (we show this in the lecture on Gaussian integrals), show that the posterior has precision $\mathbf{A} = (\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})/\sigma^2$ and mean $(\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})^{-1}\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{y}$. What do you recognise?

$$\begin{aligned}\log f(\boldsymbol{w} \mid \mathcal{D}) &= -\frac{\eta}{2\sigma^2}\|\boldsymbol{w}\|^2 - \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{w}\|^2 + \text{const} \\ &= -\frac{1}{2}\boldsymbol{w}^\mathsf{T}\frac{\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I}}{\sigma^2}\boldsymbol{w} + \boldsymbol{w}^\mathsf{T}\frac{\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{y}}{\sigma^2} + \text{const},\end{aligned}$$

so $\mathbf{A} = (\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})/\sigma^2$, $\boldsymbol{b} = \boldsymbol{\Phi}^\mathsf{T}\boldsymbol{y}/\sigma^2$ and the mean is $\mathbf{A}^{-1}\boldsymbol{b} = (\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})^{-1}\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{y}$: ridge regression. The regulariser is the prior, and as in part (a) precisions add: each data point adds $\boldsymbol{\phi}_k\boldsymbol{\phi}_k^\mathsf{T}/\sigma^2$. The posterior also gives error bars on the weights, which ridge regression alone does not.

[6 marks]

(e) Fit a line $y = w_1 + w_2 x$ ($\boldsymbol{\phi}_k = (1, x_k)^\mathsf{T}$) to the points $(0,1.1)$, $(1,2.9)$, $(2,5.2)$, $(3,6.8)$ with $\sigma = 1$ and $\eta = 0.5$. In Python, update the posterior one point at a time, by adding to the precision and to $\boldsymbol{b}$, and print the posterior mean and the standard deviations of w_1 and w_2 after each point. Check the final mean against ridge regression.

```
x = np.array([0.0, 1.0, 2.0, 3.0]); y = np.array([1.1,
2.9, 5.2, 6.8])
sig2, eta = 1.0, 0.5
A = (eta/sig2) * np.eye(2)    # prior precision
b = np.zeros(2)
for xk, yk in zip(x, y):
    phi = np.array([1.0, xk])
    A = A + np.outer(phi, phi)/sig2
    b = b + phi*yk/sig2
    m = np.linalg.solve(A, b)    # posterior mean
    sd = np.sqrt(np.diag(np.linalg.inv(A)))
    print(xk, m, sd)
Phi = np.column_stack([np.ones(4), x])
print(np.linalg.solve(Phi.T @ Phi + eta*np.eye(2), Phi.T @
y))
```

After the four points the posterior mean is $(1.02,1.90)$, exactly the ridge solution, with standard deviations 0.70 and 0.39. After one point (at $x = 0$) the slope is still 0 with its prior standard deviation $\sqrt{2} \approx 1.41$: a single point says nothing about the slope. Each new point shrinks the uncertainty.

[7 marks]

End of question 4